

MID-TERM EXAMINATION

2022

CO 216 FORMAL LANGUAGES AND AUTOMATA

Max. Marks: 40

Max Time: 2hr

Write your Roll No and Name clearly on the answer script

1. State TRUE or FALSE of the following statements:

1X10 = 10

- For every Context-Free Language, there exists a DFA to accept it. **T**
- Push-Down Automata is equal in power with Turing Machine as all languages accepted by Push-Down Automata are also accepted by a Turing Machine. **F**
- The language $L = \{a^n b^n c^n, \Sigma = \{a, b, c\}, n \geq 0\}$ is a Context-Free Language. **T**
- The language $L = \{ww^R \mid w \in \{a, b\}^+\}$ can be accepted by a Deterministic Push-Down Automata. **F**
- The language $L = \{a^n b^n, \Sigma = \{a, b\}, n \geq 0\}$ can be accepted by a Turing Machine. **T**
- Deterministic and Non-Deterministic Push-Down Automata are equivalent. **F**
- The right-most and the left-most derivation yield two parse trees for the same string belonging to a context-free language. **F**
- Regular languages are closed under the difference operation. **F**
- Regular languages are not closed under symmetric difference operation.
- Distinguishable states in a DFA are equivalent.

2. Fill up the blanks:

1X5 = 5

- Deterministic Context-Free Language class is a _____ of Context-Free Language class.
- In case of _____ grammar leftmost and _____ derivations are the same.
- Every leaf of a derivation tree is either a _____ or _____.
- A variable is useless if either cannot be _____ from the start symbol or it cannot _____ a terminal string.
- A variable is nullable if _____ can be _____ from the variable.

3. Answer the following in short

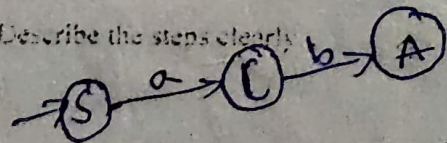
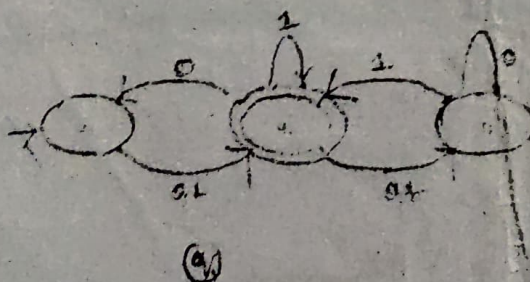
4X5 = 20

- Design a Context-Free Grammar for the language $L = \{a^m b^n, \Sigma = \{a, b\}, m, n \geq 0\}$.
- Convert the following left-linear grammar to its equivalent right-linear grammar.

$A \rightarrow Ba, Ab \mid B \rightarrow Ca, Bb \mid C \rightarrow Aa, Cb$

- Convert the following Right-Linear grammar to a DFA: $S \rightarrow abA \mid A \rightarrow baB, B \rightarrow aa \mid bb$.
- Eliminate all useless productions, if any, from the grammar $S \rightarrow aS \mid AB \mid A \rightarrow bA \mid B \rightarrow aA$.
- Design a regular grammar for the language over $\{a, b\}$ $L = \{w \mid n(w) \text{ and } m(w) \text{ are both even}\}$.

4. Convert the given NFA into its equivalent DFA. Describe the steps clearly.



$S \rightarrow asb$
 $S \rightarrow aasbb$
 $S \rightarrow aaasbbb$

$S \rightarrow asb$
 $S \rightarrow aasbb$
 $S \rightarrow asbbb$
 $S \rightarrow \epsilon$

$m = n$
 $m > n$
 $n > m$

$a^n b^n c^n$
 ww^R